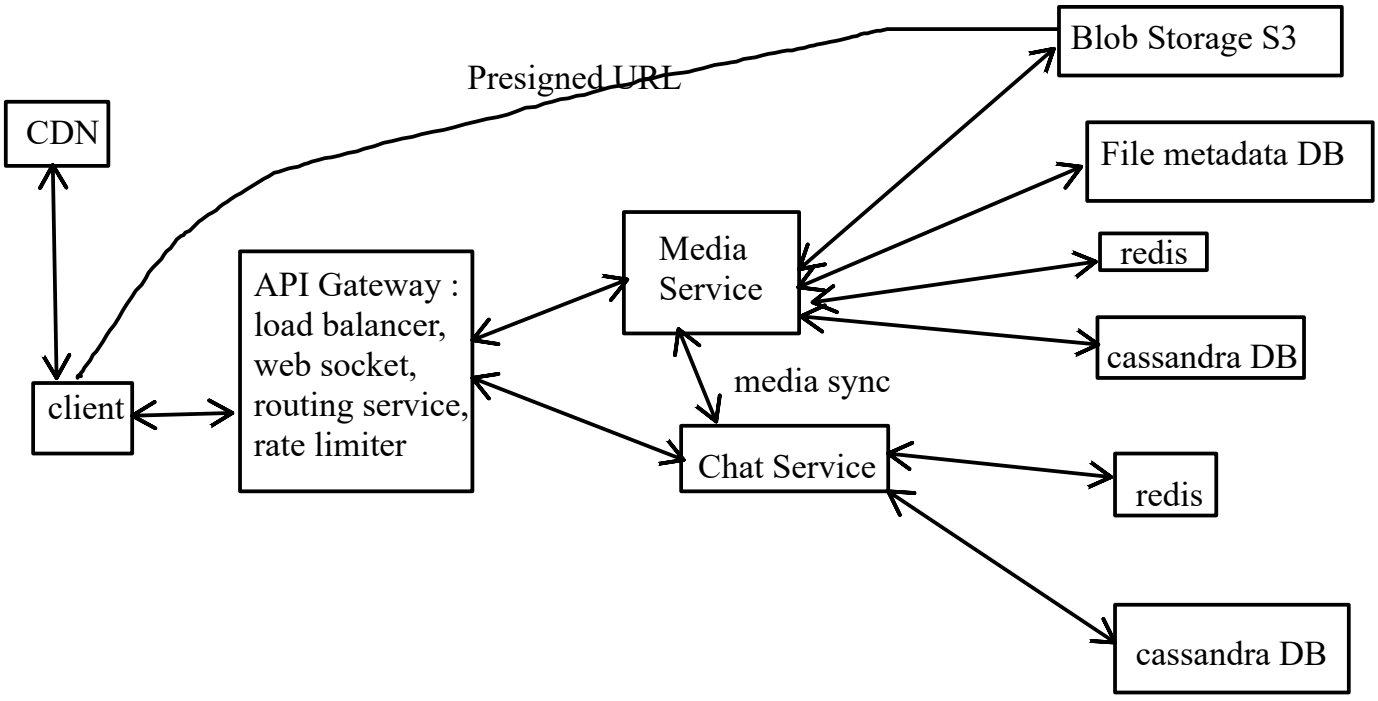
WhatsApp

Functional requirements:
user should be able to message to any other person,
user should be able to share media based on his preference,
user should be able to search for any chat based on his preference

Non-functional requirements:
data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (Here eventual consistency is fine , but here we need 99.99% uptime i.e Availability>>Consistency)

Core Entities:
user,
chat,
history,

API :
POST:/upload/media
{
userID,
{filemetadata}
}

GET:/search/{chatID}

HLD :